Validating TreeMap Vegetation Data Using Field Measurements from Talladega National Forest

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Introduction

- **TreeMap** is an imputed, tree-level forest model developed by the U.S. Forest Service. It provides wall-to-wall raster of tree species, diameter, and height across the contiguous
- U.S. Each 30×30 m pixel represents a modeled cluster of trees based on Forest Inventory and Analysis (FIA) plot data.
- This study validates TreeMap 2016 predictions (diameter, height, basal area, volume) using 2022 field measurements from 255 FIA plots in **Talladega National Forest**, to assess its suitability for operational forest management.

Data

TreeMap 2016 (Figure 1)	Field Measurements (Figure 2)
LiDAR-derived	FIA-style ground plots
Tree-level data (species, diameter, height) at 30 × 30 m resolution	255 plots (1/10 acre each) with detailed tree observations

Table 1. Comparison of TreeMap and Field Data Characteristics

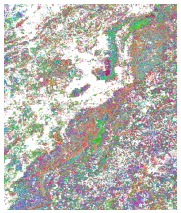

Figure 1. TreeMap 2016 vegetation raster overlay

Figure 2. GPS-based plot locations from field survey

Methodology: Matching Pixels to FIA Plots

1. Each 2022 field plot overlaps multiple TreeMap 2016 pixels (shown in different colors in Figure 3), making direct comparison difficult.
2. To spatially align the datasets, tree locations from the 2022 field measurements were mapped using azimuth and distance from the plot center, and a **convex hull** (yellow polygon) was generated to represent each plot boundary.

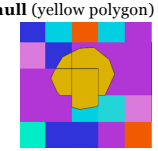

Figure 3. Example of a convex hull (yellow) intersecting multiple TreeMap pixels (colored)

3. For each overlapping TreeMap pixel, we calculated the **proportion** of its area that falls within the plot boundary, as defined by the equation below:

$$\text{Pixel Area Proportion} = \frac{\text{Area of pixel inside plot}}{\text{Total plot area (convex hull)}}$$

4. Weighted Aggregation Using Area Proportion

TreeMap estimates of diameter (DIA) and height (HT) were aggregated to the plot level using area-weighted averages:

$$\text{Weighted DIA} = \sum(\text{Pixel DIA} \times \text{Area Proportion})$$

$$\text{Weighted HT} = \sum(\text{Pixel HT} \times \text{Area Proportion})$$

5. Basal Area and Volume Estimation

Basal Area (BA) and Volume (VOL) were derived using standard forestry equations:

- **Basal Area (BA)**: $BA = 0.005454 \times \text{DIA}^2$
- **Volume (VOL)**: $VOL = 0.08168 \times \text{DIA}^{1.15697} \times \text{HT}^{0.06573}$
- For both TreeMap and field data, plot-level values were computed by summing tree-level estimates multiplied by Trees Per Acre.

TAKE-HOME MESSAGE: TreeMap 2016 significantly underestimates key forest metrics—diameter, height, basal area, and volume—limiting its accuracy at the plot level.

Results

- **Accuracy Metrics**

Prediction accuracy was assessed by comparing TreeMap 2016 estimates with 2022 field measurements for diameter, height, basal area (BA), and volume.

variable	RMSE	MAE
Diameter (inches)	3.29	2.47
Height (feet)	13.90	11.11
Basal Area (ft²/ac)	45.80	36.50
Volume (ft³)	125.62	98.89

Table 2. Prediction Accuracy of TreeMap 2016 Compared to 2022 Field Data

1.Diameter & Height: Both showed very weak correlations with field data (R^2 = 0.0045 for DIA and 0.0281 for HT).

2. Basal Area: RMSE = 45.90 ft²/ac; R^2 = 0.1113. TreeMap consistently underestimated basal area.

3. Volume: RMSE = 125.62 ft³/ac; R^2 = 0.0812. While TreeMap volume estimates reflected general magnitude, they exhibited high variability and systematic bias at the stand level.

Overall: All TreeMap variables showed weak correlations with field-measured data, indicating limited reliability for fine-scale analysis. It suggests that TreeMap requires local calibration for accurate stand-level assessment.

- **Cumulative Distribution Function (CDF) Comparison**

Figure 4 displays CDF curves for diameter, height, basal area, and volume. TreeMap 2016 predictions (blue) consistently **underestimate** 2022 field data (red), with a clear **leftward shift** indicating systematic bias—especially in the **upper quantiles**, where larger trees and denser stands are poorly represented.

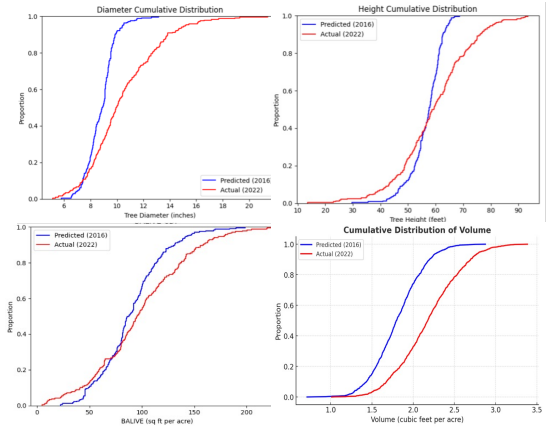

Figure 4. CDF Comparison of Tree Attributes: TreeMap 2016 vs. Field Data 2022

Conclusion & Future Work

Underestimation: TreeMap 2016 consistently underestimated all forest metrics, particularly in the upper quantiles.

Potential Cause: Temporal mismatch—growth between 2016 and 2022 may explain discrepancies.

Implication: TreeMap's current estimates may not support applications that demand stand-level precision, such as operational planning.

Future Work: Compare field measurements with FIA data closest to 2022 to reduce temporal gaps and improve validation accuracy.

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